

Stage 1: Data Processing & Feature Engineering

Raw Network Flows
CIC-IDS2017 / UNSW-NB15 /
CIC-IDS2018

Preprocessing
- Numeric columns only
- Remove null/infinite values
- StandardScaler normalization

SMOTE Balancing
k=5 nearest neighbors

Feature Selection
SelectKBest (k=30)

Train-Test Split
80% Training / 20% Testing

Stage 2: Parallel Base Learners

KNN
k=5

Random Forest
n=100
max_depth=15

XGBoost
n=50
learning_rate=0.1
depth=6

Stage 3: Weighted Soft Voting Ensemble

Weighted Soft Voting
KNN = 0.2
RF = 0.4
XGB = 0.4

Stage 4: Prediction & Evaluation

Final Prediction
Normal / Attack

Evaluation Metrics
Accuracy, Precision, Recall
F1-score, ROC-AUC

